



PROJECT UPDATE REPORT-4

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Project Update Report

AI-Powered E-Commerce Platform with Intelligent Business Insights

1. Introduction

This report summarizes the development progress of the fourth project update for our AI-Powered E-Commerce Platform with Intelligent Business Insights. The main objective of this phase was to combine the latest frontend and backend versions into one working system and extend the platform with two separate AI chatbot interfaces.

The team reviewed the latest code submitted by the frontend and backend members, identified missing integration work, and rebuilt the chatbot connection without replacing their existing store features. The completed system now connects the React interface, Django REST Framework, the SQLite database, and the local Ollama language model through tested API endpoints.

2. Frontend Development and Integration Progress

The latest React frontend retained its API calls for login, registration, products, categories, cart, checkout, orders, and the customer chatbot. Product and order pages were updated to request data from the Django backend instead of relying only on temporary frontend data. The login page stores access and refresh tokens so authenticated requests can be sent to protected endpoints.

The customer chatbot page was improved with loading feedback, disabled controls while a response is being generated, validation for empty input, and clearer error messages when Django or Ollama is unavailable. A separate AdminChat page was added with suggested business questions, a dedicated interface, and an authorization token attached to every request.

The application routes and navigation bar were updated so users can access the customer AI assistant and administrators can open the private business assistant. The frontend was tested against the new backend endpoints and both chatbot replies were displayed correctly in the browser.

3. Backend Development Using Django REST Framework

The latest backend already contained Django REST Framework view sets for users, products, categories, carts, orders, payments, and saved chat records. A new chatbot application was added without changing Rahat's existing chat application. This kept the database chat models separate from the code that communicates with the AI model.

The customer chatbot endpoint was implemented with Django REST Framework APIView, serializer-based input checking, and Response objects. It receives a customer message through POST `/api/ai/chat/`, reads active products from SQLite, sends the message and safe product information to the local phi3:mini model through Ollama, and returns the generated reply as JSON.

4. Dual AI Chatbot Implementation

The customer chatbot now answers using the real product catalogue instead of guessing product names or prices. It can explain available products, categories, descriptions, prices, and stock while being prevented from receiving user, payment, or private order information. The endpoint remains publicly available because it is intended for customer support.

A second endpoint, POST /api/ai/admin-chat/, was created for the Admin Manager chatbot. It uses the same phi3:mini model but receives a different instruction set and a larger database summary. The summary includes product counts, active products, low-stock items, users, orders, order statuses, recent orders, recorded sales values, and top-selling products when order-item data is available.

The admin endpoint requires a valid JSON Web Token and verifies that the logged-in account is a staff or administrator account. Normal users and unauthenticated visitors are rejected. The chatbot is deliberately read-only: it can explain database information but cannot create, delete, or modify products, stock, users, or orders.

5. Testing, Startup and Repository Update

Both endpoints were tested directly through PowerShell and through the React pages. Tests confirmed successful message submission, Ollama response generation, frontend rendering, administrator-only access, and separation between public product information and private business information. The customer chatbot and admin chatbot were also retested after a complete server restart.

A Windows batch launcher named Start-BestCommerce.bat was created to activate the backend environment, start Django, start the Vite frontend, and open the chatbot page automatically. The latest combined frontend and backend were uploaded to the FarhanRafid GitHub branch, while the main branch and the individual branches of other members were left unchanged. A partsChanged.md file was also added to document the integration work.

6. Current Project Status and Future Work

At the completion of this update, the project has a working React frontend, Django REST Framework backend, JWT login, database-connected customer chatbot, protected admin chatbot, and one-click local startup. Both chatbots use the same local model but receive different information and permissions according to their purpose.

The next phase will focus on improving data accuracy, adding token refresh and protected frontend routes, connecting remaining profile and checkout features, and extending the planned human-support takeover system. Additional work will include cleaner environment configuration, stronger error handling, and preparation of the one-minute demonstration video and final project documentation.